1. $r(u,v) = (\cos u, \sin u, v)$

$r_u(u,v) = (-\sin u, \cos u, 0)$ $\qquad$ $r_v(u,v) = (0,0,1)$

$$r_u \times r_v = \begin{vmatrix} i & j & k \\ -\sin u & \cos u & 0 \\ 0 & 0 & 1 \end{vmatrix} = (\cos u, \sin u, 0)$$

$\| r_u \times r_v \| = 1$

$$\vec{N} = \frac{r_u \times r_v}{\| r_u \times r_v \|} = (\cos u, \sin u, 0)$$ 高斯映射

$\vec{N_u}(u,v) = (-\sin u, \cos u, 0)$

$\vec{N_v}(u,v) = (0,0,0)$

$$\begin{pmatrix} \vec{N_u} \\ \vec{N_v} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} r_u \\ r_v \end{pmatrix}$$

2. $r(u,v) = (u, v, u^2 - \frac{v^2}{4})$

$r_u(u,v) = (1, 0, 2u)$ $\qquad$ $r_v(u,v) = (0, 1, -\frac{v}{2})$

$$r_u \times r_v = \begin{vmatrix} i & j & k \\ 1 & 0 & 2u \\ 0 & 1 & -\frac{v}{2} \end{vmatrix} = (-2u, \frac{v}{2}, 1)$$

$\| r_u \times r_v \| = \sqrt{1 + 4u^2 + \frac{v^2}{4}}$

$$\vec{N}(u,v) = \frac{(-2u, \frac{v}{2}, 1)}{\sqrt{1+4u^2+\frac{v^2}{4}}}$$

$$\vec{N_u}(u,v) = \frac{(-2,0,0)\sqrt{1+4u^2+\frac{v^2}{4}} - (-2u, \frac{v}{2}, 1)\frac{1}{2}\, 8u (1+4u^2+\frac{v^2}{4})^{-\frac{1}{2}}}{(1+4u^2+\frac{v^2}{4})}$$

$$= \frac{(-2,0,0)(1+4u^2+\frac{v^2}{4}) - (-2u, \frac{v}{2}, 1)\cdot 4u}{(1+4u^2+\frac{v^2}{4})^{\frac{3}{2}}}$$

$$= \frac{(-2-\frac{v^2}{2}, -2uv, -4u)}{(1+4u^2+\frac{v^2}{4})^{\frac{3}{2}}}$$

$$\vec{N_v}(u,v) = \frac{(0,\frac{1}{2},0)\sqrt{1+4u^2+\frac{v^2}{4}} - (-2u,\frac{v}{2},1)\frac{1}{2}\cdot\frac{1}{2}v\,(1+4u^2+\frac{v^2}{4})^{-\frac{1}{2}}}{1+4u^2+\frac{v^2}{4}}$$

$$= \frac{(\frac{uv}{2},\ \frac{1}{2}+2u^2,\ -\frac{v}{4})}{(1+4u^2+\frac{v^2}{4})^{\frac{3}{2}}}$$

$$\begin{pmatrix} \vec{N_u} \\ \vec{N_v} \end{pmatrix} = \frac{1}{(1+4u^2+\frac{v^2}{4})^{\frac{3}{2}}} \begin{pmatrix} -\frac{v^2}{2}-2 & -2uv \\ \frac{uv}{2} & 2u^2+\frac{1}{2} \end{pmatrix} \begin{pmatrix} r_u \\ r_v \end{pmatrix}$$

3. (a). $r(u,v) = (\cos u, \sin u, v)$

$K = 0$ “II”定号

(b). $r(u,v) = (u, v, u^3)$

$r_u = (1, 0, 3u^2)$ $r_v = (0, 1, 0)$

$$r_u \times r_v = \begin{vmatrix} i & j & k \\ 1 & 0 & 3u^2 \\ 0 & 1 & 0 \end{vmatrix} = (-3u^2, 0, 1)$$

$$\vec{N} = \frac{(-3u^2, 0, 1)}{\sqrt{1+9u^4}}$$

$r_{uu} = (0, 0, 6u)$ $r_{uv} = (0,0,0)$ $r_{vv} = (0,0,0)$

$$L = \frac{(0,0,6u)}{\sqrt{1+9u^4}} \qquad N = M = 0$$

$X = (0,0,0)$. $K = 0$ $II = \begin{pmatrix} L & M \\ M & N \end{pmatrix}$ 不定矩阵

4. 选择连续定向

$r(u,v) = (u\cos v, u\sin v, u)$. $u > 0$. $v \in [0, 2\pi)$

$r_u = (\cos v, \sin v, 1)$ $r_v = (-u\sin v, u\cos v, 0)$.

$$r_u \times r_v = \begin{vmatrix} i & j & k \\ \cos v & \sin v & 1 \\ -u\sin v & u\cos v & 0 \end{vmatrix} = (-u\cos v, -u\sin v, u).$$

$\|r_u \times r_v\| = \sqrt{2}\,u$

$\vec{N}(u,v) = \frac{1}{\sqrt{2}}(-\cos v, -\sin v, 1)$

单位次法向量集

在锥面顶点处，顶点处的越来越小的去心邻域内一族 C' 逼近这个去心邻域的光滑正则曲面的单位法向量.

$\vec{N}$: 顶点 $\longrightarrow$ 单位次法向量集

$\left\{ \frac{1}{\sqrt{2}}(-\cos v, -\sin v, 1) \mid v \in [0, 2\pi) \right\}$.

5. $x = 2$